Ecosystem engineers

Niche constructors have been described as “ecosystem engineers,” a term coined in 1994 by scientists Clive Jones, John Lawton, and Moshe Shachak. They outlined two kinds of ecosystem engineers. The first, allogenic ecosystem engineers, change physical materials. Take, for example, beavers building dams, woodpeckers excavating nest holes, and people mining for gravel; these activities modify the availability of resources for other species. When woodpeckers abandon their holes, smaller birds and other animals move in. If water floods a gravel pit, ducks and dragonflies can colonize it.

Other ecosystem engineers are autogenic, which means that simply by growing, they provide new habitats for other plants and animals. A mature oak tree, for example, is a suitable environment for a broader range of insects, birds, and small mammals than an oak sapling. Likewise, a coral reef provides homes for more fish and crustaceans as it grows larger.